

Figure 1: Opera extension manager interface to load/unload extensions, with Developer Mode enabled



Figure 2: Our extension running in the browser — highlight the text, right click on it, select 'Privacy Search on...' and any one of the sub-menu options from the context menu

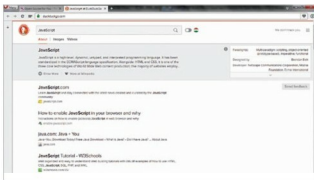


Figure 3: Extension providing results for the highlighted text in the new tab of the Opera Web browser

some function. In our case, we are creating a parent menu called 'Privacy Search on...' using the first function, and sub-menu options called 'Search on DuckDuckGo for:', 'Search on Encrypted Google for:' and 'Search on Ixquick for:' using the next three functions.

When users click or select one of these sub-menu options, a specified function is called. For example, when users click the 'Search on DuckDuckGo for:' option, the `searchTextDuckDuckGo()` function is called. The 'searchTextDuckDuckGo' function gets the value of the element 'info.selectionText'; it is appended with the URL of the DuckDuckGo search engine and is stored in a

variable named `myQuery`, which then loads in a new tab with the search results.

That's it with the coding part; now let us load and test the extension.

Loading and installing the extension

In the Opera Web browser, extensions are packaged as `.nrx` files. You can do this by going through the following steps.

Type 'opera:extensions' in the address bar of an Opera browser and press `Enter` or the `Ctrl+Shift+e` keys.

Click the 'Developer Mode' option located on the top right to enable it. By doing this, you get the ability to inspect various parts of the extension using the browser's developer tools. Besides, you can make changes to your extension's code and also preview the effects by using the reload button.

You will get two options, 'Load unpacked extension' and 'Pack Extension', as shown in Figure 1.

With the 'Load unpacked extension' option, you can navigate to the directory where your extension files are, and select that extension folder. This loads the extension.

With the 'Pack extension' option, you can select your extension folder and package it in a `.nrx` file.

After creating the `.nrx` file, just drag and drop it in the Opera browser to install it.

Running the extension in the Opera Web browser

After installing/loading the extension, you can test it by highlighting some text and then selecting the 'Privacy Search on...' option and any one of the sub-menu options, from the right click context menu. For example, we tested the extension by highlighting the word 'JavaScript' on <http://opensourceforu.eftimes.com/> and then selected the 'Search on DuckDuckGo for:' sub-menu option from the 'Privacy Search on...' option in the right click context menu, as shown in Figure 2. This opened the DuckDuckGo search results for 'JavaScript' in a new tab in the Opera browser, as shown in Figure 3. **END**

References

- [1] <https://dev.opera.com/>
- [2] [https://en.wikipedia.org/wiki/Opera_\(web_browser\)](https://en.wikipedia.org/wiki/Opera_(web_browser))
- [3] <https://www.duckduckgo.com/>
- [4] <https://encrypted.google.com/>
- [5] <https://www.ixquick.com/>

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